Comparative Evaluation of SARIMA, FB Prophet, and LSTM for Hourly Demand Forecasting in Saudagar Kopi

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***Abstract*—Demand forecasting plays an important role in the food and beverage (FnB) industry by helping businesses anticipate customer demand and support operational planning during peak periods. This study conducts a comparative empirical evaluation of three forecasting models, Seasonal Autoregressive Integrated Moving Average (SARIMA), FB Prophet, and Long Short-Term Memory (LSTM) to predict hourly demand patterns in a coffee shop environment. The models were evaluated using hourly demand patterns in a coffee shop environment. The models were evaluated using hourly sales data collected from Saudagar Kopi in Jakarta over a three-month period (April-June 2025). Forecasting performance was assessed using Root Mean Squared Error (RMSE) as the primary evaluation metric. The results indicate that the LSTM model achieves the lowest forecasting error and is better able to capture abrupt demand spikes and high-frequency fluctuations compared with SARIMA and FB Prophet, which perform well in modeling trend and seasonality but tend to underestimate sudden peak-hour demand. These findings highlight the potential of deep learning approaches for modeling complex demand patterns in high-frequency time-series data. However, the results should be interpreted within the context of a limited dataset and a single business environment. Future research may explore alternative models that address zero-inflated demand patterns and evaluate forecast performance across multiple retail locations.**

Keywords—Demand Forecasting, Peak Hours, SARIMA, LSTM, FB Prophet.

I. Introduction

The coffee shop industry has expanded rapidly in recent years, driven by young consumers and urban professionals who increasingly use coffee shops as multifunctional spaces for socializing, studying, and remote working. Indonesia, the world's 4th largest coffee produce [1], has a strong coffee culture that supports this growth. Coffee, once associated with lower-class and middle-class consumption, has evolved into a modern lifestyle product with broad market appeal [2], positioning coffee shops as an important segment within the service economy.

As customer activities diversify, operational challenges arise, particularly during peak hours, when high customer traffic can lead to long queues, delayed orders, and inconsistent service quality. These issues are evident at Saudagar Kopi, a popular coffee shop located in Jakarta's Sabang business district, which attracts office workers on weekdays and general visitors on weekends. Peak-hour congestion has resulted in service delays and negative customer feedback, posing risks to satisfaction, loyalty, and business sustainability.

Demand forecasting provides a strategic approach for anticipating customer volume and optimizing operations. Accurate forecasting can support improvements in resource allocation, inventory management, pricing strategies, and promotional planning in retail and food service settings [3] In coffee shops, demand prediction may help inform capacity management and waiting times, which are key drivers of customer satisfaction and loyalty [4]

Previous studies have implemented various forecasting models including DNN (Deep Neural Networks), Autoregressive Integrated Moving Average (ARIMA), and Long Short-Term Memory (LSTM) to identify the most suitable approach based on data characteristics [5]. LSTM models have effectively captured temporal dependencies such as seasonality and day-of-week patterns [6] and are widely used for intelligent demand planning in modern supply chains [7].

This study investigates operational inefficiencies at Saudagar Kopi, a high-traffic coffee shop in Jakarta, particularly during peak hours that lead to delayed orders and customer dissatisfaction. To address this, the research proposes the use of demand forecasting to predict customer volume and support operational planning. This study examines several forecasting models to determine which approach is most suitable for the characteristics of hourly sales data. Therefore, this study conducts a comparative empirical evaluation of 3 forecasting models, SARIMA, FB Prophet, and LSTM to analyze their effectiveness in predicting hourly demand patterns in a coffee shop context.

II. LITERATURE REVIEW

A. Business Overview

Saudagar Kopi is a coffee shop located on Jl. Sabang No 26F, Central Jakarta. Established in 2012, the shop specializes in strong and bold coffee made from single-origin beans and house blends while also offering Indonesian dishes, snacks, and packaged roasted beans. Its primary customers include coffee enthusiasts and office workers seeking high-quality coffee in a comfortable environment.

Products are sold directly in-store and through delivery platforms such as GoFood, GrabFood, and ShopeeFood, while roasted beans are sold through online marketplaces. Operationally, the Purchasing & Warehouse division manages ingredient procurement and inventory, while the kitchen and bar request supplies based on daily and weekly needs. Currently, production planning relies largely on intuition rather than structured forecasting. This research focuses on the production process, particularly forecasting demand based on food and beverage sales. Figure 1 illustrates the operational business process of Saudagar Kopi, highlighting the stages where demand forecasting can support production planning and preparation during peak demand periods.

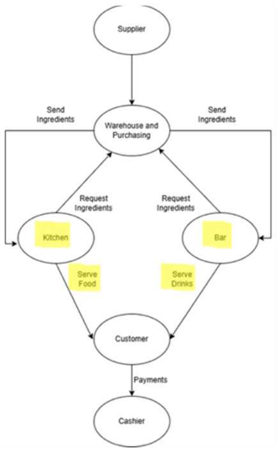

Fig. 1. Business Process Diagram of Saudagar Kopi.

B. Theoretical Foundations of Time Series Forecasting

1) Demand Forecasting

Demand Forecasting is an essential component of operational planning that enables organizations to estimate future demand for products or services. Accurate forecasts can support inventory preparation, resource allocation, and production planning particularly in industries with fluctuating demand such as food and beverage services. Recent studies have increasingly applied machine learning and deep learning approaches to improve forecasting accuracy. Models such as ARIMA, LSTM, and hybrid methods have been widely compared across different demand forecasting contexts [8] [9]. Deep learning models like LSTM have shown strong performance in forecasting complex demand patterns and short shelf-life products [10].

2) SARIMA Model

The Seasonal Autoregressive Integrated Moving Average (SARIMA) extends the ARIMA framework by incorporating seasonal components to capture recurring patterns in time-series data. The model is represented as SARIMA(p,d,q)(P,D,Q)s, allowing simultaneous modelling of non-seasonal dynamics and seasonal behavior. SARIMA has been widely applied in forecasting domains such as energy demand and economic indicators due to its ability to model trend and seasonality effectively [11]. Model performance depends on appropriate parameter identification, typically guided by statistical tools such as ACF, PACF, and information criteria [12]. For example, Ghodusinejad et al. (2025) applied SARIMA to forecast electricity and gasoline demand, demonstrating its effectiveness in identifying long-term demand trends [13].

3) LSTM

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to model sequential data by capturing long-term temporal dependencies. Its memory cell architecture allows the model to retain relevant information from previous time steps, making it suitable for time-series forecasting tasks [14]. Recent studies have demonstrated the strong forecasting capabilities of LSTM-based models. For instance, Yingnan and Xiaowei (2025) showed that an ES-LSTM model achieved lower forecasting errors compared with traditional approaches such as LSSVM and BPNN highlighting the effectiveness of LSTM in modelling dynamic demand patterns [15].

4) FB Prophet

FB Prophet is a forecasting model developed by Facebook to handle time series data with complex seasonality, trend changes, and irregular events. The model decomposes time series data into trend, seasonal, and holiday components, allowing it to capture multiple seasonal patterns simultaneously [16]. Prophet uses a Bayesian framework for parameter estimation and can handle missing data, trend shifts, and outliers effectively. In retail and food service forecasting contexts, Prophet has demonstrated competitive performance compared with traditional statistical models [17]

C. Related Works

Previous studies have applied various statistical and machine learning approaches to demand forecasting across multiple domains. For example, Elshewey et al. proposed a hybrid CNN-ResNet50-LSTM model for climate change forecasting that outperformed traditional regression models [18]. Aldahmani et al. introduced a deep learning traditional forecasting model combining BiLSTM and NARX to improve supply chain demand prediction [19]. Other studies have explored hybrid forecasting techniques and clustering-based approaches to manage intermittent demand in inventory systems [4].

In the food and beverage sector, researchers have also explored forecasting techniques using statistical models. For example, Alzami et al. applied SARIMAX and VAR models with weather data to forecast demand in F&B SMEs, while Thatphet and Ruangchoengchum used ARIMA to identify peak hours in a coffee shop environment [20]. However, many of these studies focus on broader supply chain or inventory management contexts rather than high-frequency demand fluctuations within service-oriented businesses. Despite extensive research on demand forecasting methods, limited studies have examined the comparative performance of statistical, additive regression, and deep learning models using high-frequency hourly data in coffee shop environments. This study addresses this gap by comparing SARIMA, FB Prophet, and LSTM models to forecast hourly demand patterns in a coffee shop setting, enabling the identification of peak periods and supporting operational planning.

III. Methodology

A. Data Understanding

The objective of this study is to develop a demand forecasting model for food and beverage products at Saudagar Kopi to improve product availability during peak hours. The dataset consists of hourly sales records extracted from the POS system *Solis Resto* during April – June 2025. The records include transactions date, menu item name, hourly quantities sold (00:00-25:59), hourly transaction value, and daily totals. The final dataset contains 8,903 rows and 57 columns, comprising one datetime field, 55 numerical fields, and one categorical field. These observations are structured as hourly sales data and form the basis for forecasting peak-hour demand. To evaluate forecasting performance, the dataset was divided into training and testing sets using a chronological split to preserve the temporal structure of the time series. The earlier portion of the data was used for model training, while the most recent observations were reserved for testing to evaluate out-of-sample forecasting accuracy.

B. Data Pre-Processing

The preprocessing phase focused on refining the dataset to enhance model readiness and ensure the relevance of the features to the research objective. To streamline the analysis, the study focused on Long Black the highest-volume product, as a representative proxy for forecasting rush hour trends. Irrelevant features were excluded, retaining only the transaction date, time, sequence number, and quantity (QTY). Initial EDA revealed significant zero-inflation (skewness = 2.98), with over 1,400 out of 2,184 observations containing structural zeros from non-operational hours. These were addressed through three distinct strategies. For Sarima Adopted a data-reduction approach by isolating only active operational hours and narrowing the scope to peak rush-hour windows. FB Prophet Retained the full operational timeline but mitigated the impact of zeros using binary exogenous indicators (is_rush_hour, is_lunch_rush). And for LSTM Leveraged temporal encoding and demand occurrence features to help the network navigate sparsity.

Outliers were identified using the Interquartile Range (IQR) method. However, a cross-check revealed that these spikes were not data entry errors but were driven by external factors such as holidays, weekends, and special events. Consequently, these outliers were retained to preserve the authentic demand volatility. To account for these variances, exogenous variables (e.g., holiday indicators and weekend flags) were integrated into the model to improve forecasting accuracy during high-traffic periods.

C. Model

To evaluate the models' predictive accuracy, the dataset was partitioned into training and testing sets using a 24×7 (168 hours) split. This window represents a full one-week cycle, ensuring the model is validated against every day of the week. This accounts for the distinct variance between weekday and weekend consumer behavior, providing a robust test for a coffee shop's operational cycle.

1) SARIMA

A distinct approach was implemented for the Seasonal Autoregressive Integrated Moving Average (SARIMA) model. Rather than modeling the full 24-hour cycle, the analysis was narrowed specifically to identified rush-hour windows by isolating peak periods, the model avoids the data sparsity and low-variance noise typical of non-peak hours. Within this focused scope, day_of_week and hour were added as exogenous variable. It allowed the SARIMA parameters to more accurately capture the high-intensity fluctuations and volatility inherent in peak coffee demand without being biased by off-peak lulls.

2) FB Prophet

In the Prophet framework, the model was configured to move beyond standard seasonal decomposition by incorporating domain-specific exogenous variables In addition to Prophet's built in daily and weekly seasonality, we added binary indicators for is_rush_hour and is_lunch_rush, as well as a categorical hour regressor. These allow the model to specifically penalize or boost predictions during known high-volatility periods that a standard Fourier-based seasonality curve might smooth over. Yearly seasonality was disabled. The changepoint_prior_scale (tested at 0.1 and 0.5) and seasonality_mode (Additive vs. Multiplicative) were evaluated to determine how these exogenous surges interact with the baseline trend.

3) LSTM

To capture non-linear dependencies and long-term patterns, a Stacked LSTM network was developed using a robust suite of engineered exogenous features:

a. Feature Engineering: To represent temporal periodicity, hour and day_of_week were transformed via cyclical sine/cosine encoding. The model's inputs were further enriched with lag features and rolling window statistics (mean/variance) to capture momentum, alongside binary flags for is_rush_hour, is_lunch_rush, and holidays.
b. Architecture & Optimization: The network employs two stacked LSTM layers with tanh activations for hidden states and ReLU for the final dense layer. Critical hyperparameters including hidden units, dropout, and learning rates were tuned using Bayesian Optimization.
c. Training Dynamics: The model was trained on 24 hour sequences using the Adam optimizer and Huber Loss, the latter chosen for its robustness against demand outliers. To ensure generalization, Early Stopping was implemented with a 10 epoch patience window, while the target variable was scaled to improve convergence

D. Evaluation

The primary performance benchmark for all models is Root Mean Squared Error (RMSE). RMSE was selected over MAE and MAPE because it provides a quadratic penalty for large errors. In the food and beverage industry, a significant underestimation of demand during peak hours is operationally more costly than small off-peak deviations. By using RMSE, we ensure the models are optimized to mitigate the highest operational risks.

IV. Result and Discussion

A. Exploratory Data Analysis

Exploratory analysis identified the five best-selling items and revealed distinct temporal demand patterns. Several products (e.g., Long Black, Roti Srikaya, and Singkong Garlic) peak during morning hours, while others (e.g., Perfect Mineral Water, Pisang Tempura) Weekly patterns show higher demand during mid-month, and hourly trends indicate two dominant peaks: 07:00-07:59 and 12:00-12:59.

Timestamp validation confirmed consistent one-hour intervals. The distribution was right-skewed due to frequent zero demand hours, but this does not affect the SARIMA since normality is required only for residuals. The presence of numerous zero-demand hours reflects typical retail and food-service transaction patterns, where sales do not occur continuously throughout the day. Such zero-inflated observations may influence forecasting performance, as statistical models like SARIMA and FB Prophet tend to produce smoother forecasts that may underestimate sudden increases in demand following periods with no transactions. Outlier analysis identified occasional spikes (3-13 units), typical of food-service demand variability. Seasonal decomposition showed clear trends, weekly seasonality, and irregular components. Seasonal differencing achieved stationarity, supporting SARIMA model requirements.

B. Model Evaluation

1) SARIMA Model

TABLE I. SARIMA result for the Morning Rush Period (07:00–08:00) and the Lunch Rush Period (12:00–13:00)

SARIMA RESULT				
Period	***MAE***	***MSE***	***RMSE***	***MAPE***
Morning (07:00–08:00)	0.3617	0.2772	0.5265	24.91%
Lunch (12:00–13:00)	0.4176	0.2959	0.5440	32.29%

During the morning rush hour (07:00-08:00), SARIMA achieved RMSE = 0.5265 and closely followed the observed trend. For lunchtime (12:00-13:00), the model achieved RMSE = 0.544, with accurate seasonality but weaker performance on sharp spikes. Residual diagnostics indicated no autocorrelation and acceptable model adequacy.

2) FB Prophet Model

Prophet, enhanced with rush-hour indicators, achieved RMSE = 0.46, outperforming SARIMA. The model captured weekly and daily seasonality and produced stable forecasts. However, Prophet consistently underestimated peak-hour spikes due to its smoother additive structure.

3) LSTM Model

The LSTM model delivered the best performance with RMSE = 0.1535, substantially lower than the statistical models. Predictions aligned closely with actual values, including sudden demand increases. Residuals were symmetrically distributed, and scatter plots demonstrated strong correlation with observed data. LSTM effectively captured both temporal patterns and high-frequency variability.

C. Interpretation of Results

The comparative analysis highlights that SARIMA is effective in capturing trend and seasonality but struggles with sudden volatility. Prophet produces smooth and interpretable forecasts but tends to underestimate peak-hour spikes. In contrast, LSTM achieved the lowest error rate and provided the most robust forecasts, effectively modeling high-frequency fluctuations in hourly sales. This improved performance can be attributed to the ability of LSTM networks to capture nonlinear temporal dependencies and complex sequential patterns in time series data. Unlike SARIMA and FB Prophet, which rely on linear relationships and smoother seasonal components, LSTM can learn irregular demand spikes and short-term fluctuations that frequently occur during rush-hour periods in food service environments. The substantial reduction in RMSE compared with the statistical models also indicates a meaningful improvement in forecasting accuracy for high-frequency hourly demand patterns.

D. Research Gap

While prior studies, such as Alzami et al. [20], employed SARIMAX with daily and weather data for F&B demand forecasting, they did not address short-term fluctuations characteristic of rush-hour sales in coffee shops. Moreover, earlier works have relied heavily on classical statistical methods, with limited exploration of deep learning models in this context. This study addresses the gap by using hourly data from a three-month period and comparing statistical, additive regression, and deep learning models, thereby extending prior research on forecasting methods in service industries.

E. Findings and Contributions

The empirical results indicate that the LSTM model outperforms SARIMA and Prophet in forecasting short-term demand at Saudagar Kopi. While SARIMA successfully captures trend and seasonality and Prophet provides interpretable and stable forecasts, both models show limitations in representing sudden demand spikes and high-frequency fluctuations during rush hours. In contrast, the LSTM model achieves substantially lower RMSE values and is better able to track actual demand patterns, including abrupt peaks.

A key limitation of this study is the relatively short observation period of three months (April–June 2025), which restricts the ability to capture annual cycles or long-term structural changes. In addition, the dataset originates from a single coffee shop, which may limit the generalizability of the findings to other businesses or service environments. Nevertheless, the study provides several contributions. From a theoretical perspective, it enriches the literature by comparing statistical, additive regression, and deep learning approaches on high-frequency sales data, highlighting the impact of demand fluctuations on model performance. From a practical perspective, the study offers insights for managers in the food and beverage sector, suggesting that LSTM-based forecasting could support more accurate staff scheduling, inventory management, and resource allocation during peak periods.

V. Conclusion

This research examined the significance of demand forecasting, particularly within the Food and Beverage industry. Demand forecasting is a crucial instrument that enables coffee businesses to anticipate consumer preferences, hence optimizing resource allocation and reducing customer wait times. The major objective was to demand forecasting to optimize resource allocation and improve customer experience. Principal findings demonstrate that the optimal model for demand forecasting, particularly for short time frames and hourly data, is LSTM. The results indicate that LSTM performs exceptionally well, with predicted values closely aligning with actual values, whereas the SARIMA and FB Prophet models well capture trends and seasonality but underestimate unexpected peak sales during Rush Hour. The ramifications of these discoveries are significant. SARIMA and FB Prophet encounter difficulties with abrupt peaks and high-frequency swings, despite the use of feature engineering and exogenous variables. It also demonstrates that the LSTM model can manage abrupt peaks and high-frequency variations. Furthermore, the work enhances the current literature by comparing forecasting models on seasonal time series data characterized by significant fluctuations.

This research recognizes constraints, including a brief time frame, as the dataset encompasses only three months of sales and contains numerous zero values. The results may solely represent conditions within a three-month period (April – June 2025) and not the overall state of Saudagar Kopi, hence affecting the performance of SARIMA and FB Prophet.

Future study should concentrate on the application of alternative models capable of managing high-frequency swings and abrupt sales peaks. The model must be capable of managing substantial zero values in the dataset, as this is a regular occurrence in coffee shops where there is no demand (zero sales). While LSTM demonstrates exceptional performance, it is too costly for small and medium enterprises, such as coffee shops, unless the business has many locations. In future implementations, the FnB manager can utilize machine learning for demand forecasting to improve operational efficiency.

In conclusion, demand forecasting is crucial, particularly for resource allocation in the food and beverage industry. Deep learning algorithms, such as LSTM, can be employed by food and beverage managers for resource allocation to improve customer experience.

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